Luca Melfi

508-439-7281 | lmelfi@andrew.cmu.edu | https://www.linkedin.com/in/lucamelfi18/

EDUCATION

Carnegie Mellon University, Pittsburgh, PA
Bachelor of Science in Mechanical Engineering (GPA: 3.93/4.00, Engineering Dean's List, May 2027)
Master of Science in Mechanical Engineering, Concentration in Robotics and Control Systems (December 2027)

TECHNICAL SKILLS

Controls & Software: Python, C, C++, MATLAB/Simulink, OpenCV, Teensy, Arduino, Raspberry Pi, Stepper Drivers
Design & Analysis: SolidWorks, FEA (ANSYS: Thermal, Structural, Modal), DFM, GD&T, Siemens Plant Simulation, SimTalk
Prototyping & Fabrication: 3D Printing (SLA, FDM), Laser Cutting, Soldering, Manual Mill, Lathe, Waterjet, Welding

RELEVANT COURSEWORK

Robot Dynamics & Analysis, Modern Control Theory, Electromechanical Systems Design, Applied FEA, Heat Transfer

INTERNSHIP & RESEARCH EXPERIENCE

Advanced Mechanical Engineering Intern, Lila Sciences | May 2026 – August 2026
- Architected mechanical design (SolidWorks) of a 21-instrument lab automation work cell (705-line BOM, 50+ CAD models), driving it from concept layout through Final Design Review to build release
- Engineered a pneumatic distribution system, sizing regulators and filtration to deliver instrument-specific air quality
- Designed 20+ custom precision fixtures in SolidWorks, applying GD&T and tolerance stack-up analyses to accomplish first-pass acceptance across all outside vendors

Graduate Researcher, The Biohybrid and Organic Robotics Group (B.O.R.G.) | February 2025 – Present
- Designed and implemented an electromechanical linear-actuation and load-measurement system achieving 0.04 mm positional accuracy and millinewton force resolution to quantify the biomechanics of soft-bodied organisms
- Programmed an open-loop C++ stepper-motor controller with 3,200-step/rev micro-stepping for precise tissue probing, verifying positional accuracy against stylus-micrometer measurements in Python
- Simulated biomechanical properties of soft-tissue organisms by applying a neo-Hookean hyperelastic model to size silicone test-phantoms, tuning the geometry for a 3.92 N physical force calibration

Product Supply Engineering Intern, Procter & Gamble | May 2025 – August 2025
- Deployed an AI-driven machine-vision and inline-metrology system, enabling 100% automated inspection and eliminating 520 hours of manual labor annually
- Validated PLC logic and optimized throughput for automated depalletization machinery prior to installation by building interactive discrete-event simulations in SimTalk and Siemens Plant Simulation

PROJECT EXPERIENCE

Two-Link Robotic Arm | Carnegie Mellon University | May 2025
- Developed a two-link robotic arm and custom control architecture in Python, achieving <0.5" positioning accuracy by integrating encoder-based odometry feedback with a Raspberry Pi PID controller
- Implemented an autonomous motion planning pipeline in Python, utilizing inverse-kinematics and configuration-space algorithms to compute and execute collision-free trajectories in real-time

Robotic Air Hockey Table, Mechanical Lead | BUILD18 Hackathon | November 2024 – March 2026
- Integrated an OpenCV puck-tracking pipeline to command autonomous robot actuation at 1.5 m/s
- Engineered and rapid-prototyped (SolidWorks, 3D Printing) an active-cooling enclosure for a high-speed stepper motor, utilizing thermal FEA simulations to validate designs and hold peak temperatures under 65°C

LEADERSHIP & TEACHING EXPERIENCE

Undergraduate Teaching Assistant, Numerical Methods | August 2026 – Present
- Guided 100+ students in writing numerical algorithms in Python by leading recitations and office hours

AC Mellon Men's Club Soccer, President, Captain | April 2026 – Present
- Lead operations for a 40-member club, managing travel, scheduling, and $10K+ budget as President and Captain